Michael Remington Coulter

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PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer Apr 2023 - Present

Develop Interconnect Architect, Arm's Python/TypeScript SoC interconnect configuration tool for internal engineers and external customers, alongside AI tools for engineering workflows.

- Implemented Python Model Context Protocol (MCP) tools compiling typed intent into dependency-ordered execution plans, with checkpoints, semantic checks and rollback.
- Separated shared AI tools from project-specific capability packs using manifests; consolidated MCP implementation and acceptance assets under product ownership.
- Validated creation, import, modification, diagnosis and rollback in two bounded agent evaluations; rollback restored the exact configuration hash. 92 live acceptance checks passed.
- CMN GUI Lead: created and own all coherent-mesh GUI tooling from CMN-800 onward, including device/group configuration, custom routing and deadlock investigation.
- Built a Python interconnect migration engine; preserved topology, routes and placement, with 4,130 valid configurations passing design-rule checks and five curated invalid cases rejected.
- Integrated Python configuration policies with Angular editors, coordinating versioned state, checkpoints, recovery and editing permissions; invalidated compliance after design changes.
- Connected current and legacy Python backends through an authenticated sidecar with API/path restrictions and provenance checks, enabling tooling across supported engine versions.
- Built semantic design comparison across isolated backend sessions, checking topology, routes, attributes and source YAML; added visual overlays and HTML change reports.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025): multi-agent orchestration and context-gathering system.

Software Engineer Apr 2020 - Apr 2023

- Built Flask services for Jupyter notebook lifecycle and AWS EFS persistence in Pelion's embedded-ML platform; designed PostgreSQL schemas and end-to-end tests.

Graduate Software Engineer Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware to test board I/O; contributed to test-shield PCBs. Arm Quality Award (team, 2020).

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, SQL, Verilog
Architecture: MCP, typed schemas, dependency planning, topology migration, semantic validation
Applications / delivery: Angular, Electron, Flask, PostgreSQL, AWS EFS; Git, Jenkins, Playwright

EDUCATION

The University of Texas at Austin
Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits